

Birth defects

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Objectives:

- Define birth defects and enumerate their causes.
- Define major and minor anomalies.
- provide birth defects examples, and explain their clinical significance.

Birth defect

It is a condition existing at or before birth regardless of the cause.

Inherited / acquired.

Congenital abnormalities

A congenital disorder is a condition existing at birth and often before birth, or that develops during the first month of life (neonatal disease), regardless of causation.

- Major: *serious structural abnormality* has an **adverse outcome** on either functional or social acceptability of the individual
- Minor: with *small risk of serious sequel* ; are of neither medical or cosmetic importance

Incidence of Structural Abnormalities

incidence	%
Spontaneous miscarriage :	
First trimester	80-85
Second trimester	25
All babies	
Major abnormality apparent at birth	2-3
Major abnormality apparent later	2
Minor abnormality	10
Death in perinatal period	25
Death in first year of life	25
Death at 1-9 years	20
Death at 10-14 years	7.5

- **Mortality:**

The high incidence of major and minor abnormalities noted in defects observed in early **spontaneous miscarriages**

- **1st year of life**: 25% of all deaths are the result of major structural abnormalities
- **from 1 – 10 years** value falls to 20%
- **From 10 – 15 years** becomes around 7.5%
-

Major congenital structural abnormalities

- Cardiovascular : VSD, ASD, PDA, F4
- CNS : hydrocephaly, microcephaly, anencephaly, NTD
- Gastrointestinal: cleft lip/palate, esophageal atresia, imperforate anus
- Urogenital : bilateral renal agenesis, infantile polycystic kidney
- Limb : transverse amputation

CNS



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Baby with Microcephaly

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GIT



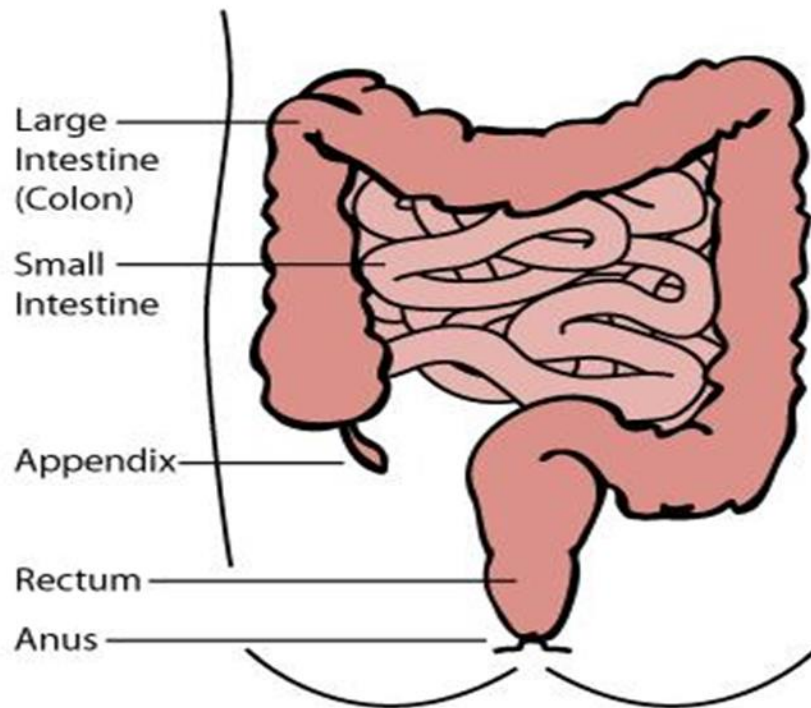
Unilateral cleft lip



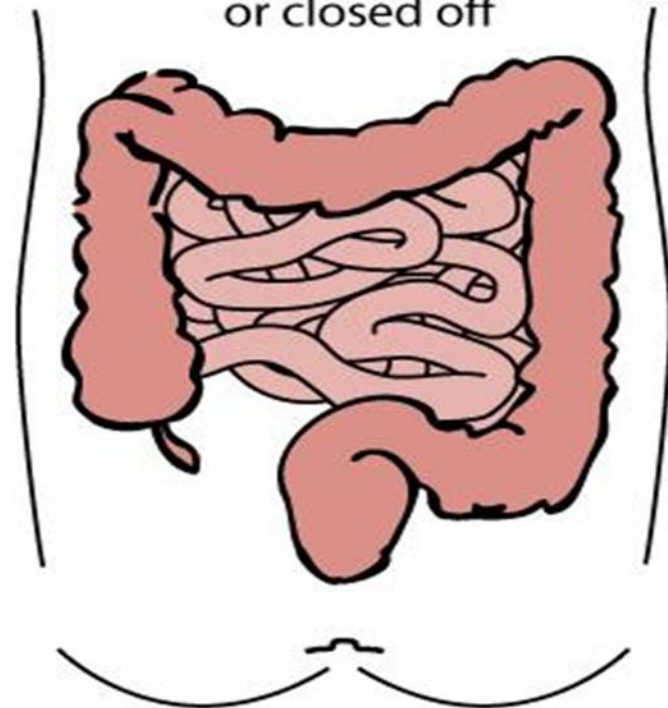
Bilateral cleft lip

FORMS OF IMPERFORATE ANUS - FRONTAL VIEW

Normal Anatomy



Anal opening missing or closed off



Examples of minor structural congenital abnormalities

- **Periauricular pit or tag**
- **Epicanthic fold**
- **Lacrimal duct stenosis**
- **Brushfield spots of the iris**
- **Lip pits**
- **Single palmar crease, 5th finger clinodactyly**
- **Syndactyly between 2nd & 3rd toes**
- **Supermeumaray nipple**

- **Epicanthic fold**
- **Syndactyly between 2nd & 3rd toes**



- **Classification and mechanisms of Birth defects**

- 1-single abnormalities

- Malformation
- Disruption
- Deformation
- Dysplasia

- 2-multiple abnormalities

- Sequence
- Syndrome
- Association

What causes birth defects?

- **Genetic:**

Chromosomal

Single gene disorders

- **Environmental factors**

- **Multifactorial :**

a combination of these factors, can cause birth defects.

single abnormalities

- **Mechanism:**

Malformation

Deformation

Desruption

Dysplasia

Classification of Birth defect

- **Malformation**

- **Arrested** or **misdirected** development of a particular tissue or organ **early in gestation**.
- **Primary** and **intrinsic**

Malformation

- **Examples:** Congenital heart defects (VSD, ASD)

Cleft lip/palate

Neural tube defect



Malformation

Causes:

Most malformations involving **only a single organ** show **multifactorial inheritance**, implying an interaction of gene(s) with other environmental factors

Multiple malformations are more likely to be due to **chromosomal abnormalities** but may be due to **single gene defect**

Malformation

Causes:

- **Chromosomal abnormalities**
- **Single gene disorder**
- **Multifactorial inheritance**

Classification of Birth defect...

- Disruption
 - Extrinsic factors that **disrupt** normal development
 - **Secondary** and **extrinsic**
 - Not genetic

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Disruption:

- Involves breakdown of normal tissues.
- Results from **external factors disrupting** the normal developmental process.

Known to be **2ry** or **extrinsic malformation**

Disruption:

- **Causes: Not genetic**
e.g. (ischemia, infection, trauma)
Amniotic band

Classification of Birth defect ...

- Deformation

Abnormal **mechanical force** which distorts an otherwise normal structure.

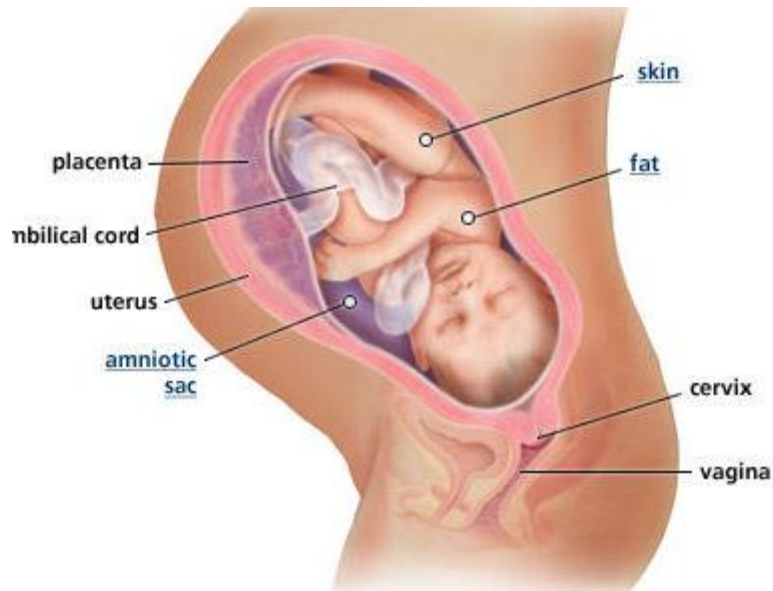
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Deformation

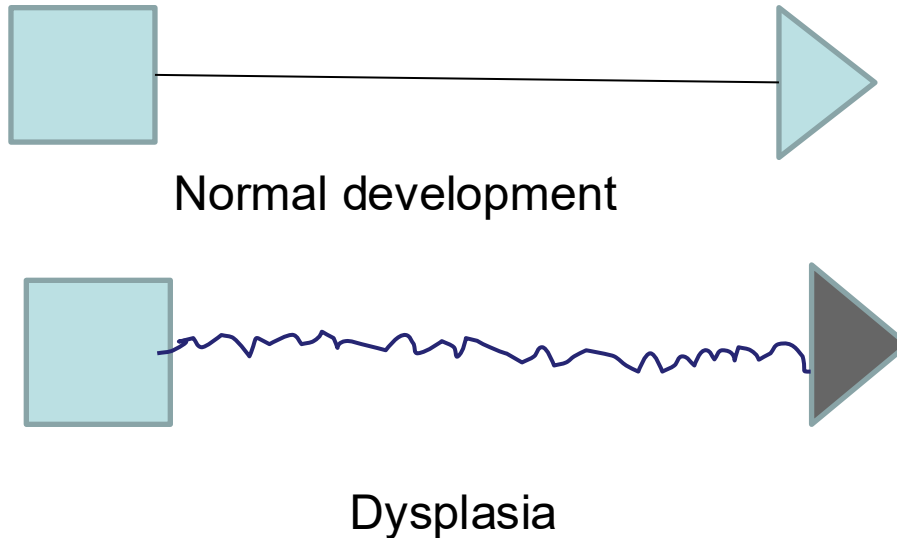
Causes:

Oligohydramnios or abnormal structure of the uterus may lead to *hip dislocation or positional talipes* (club foot)



Dysplasia

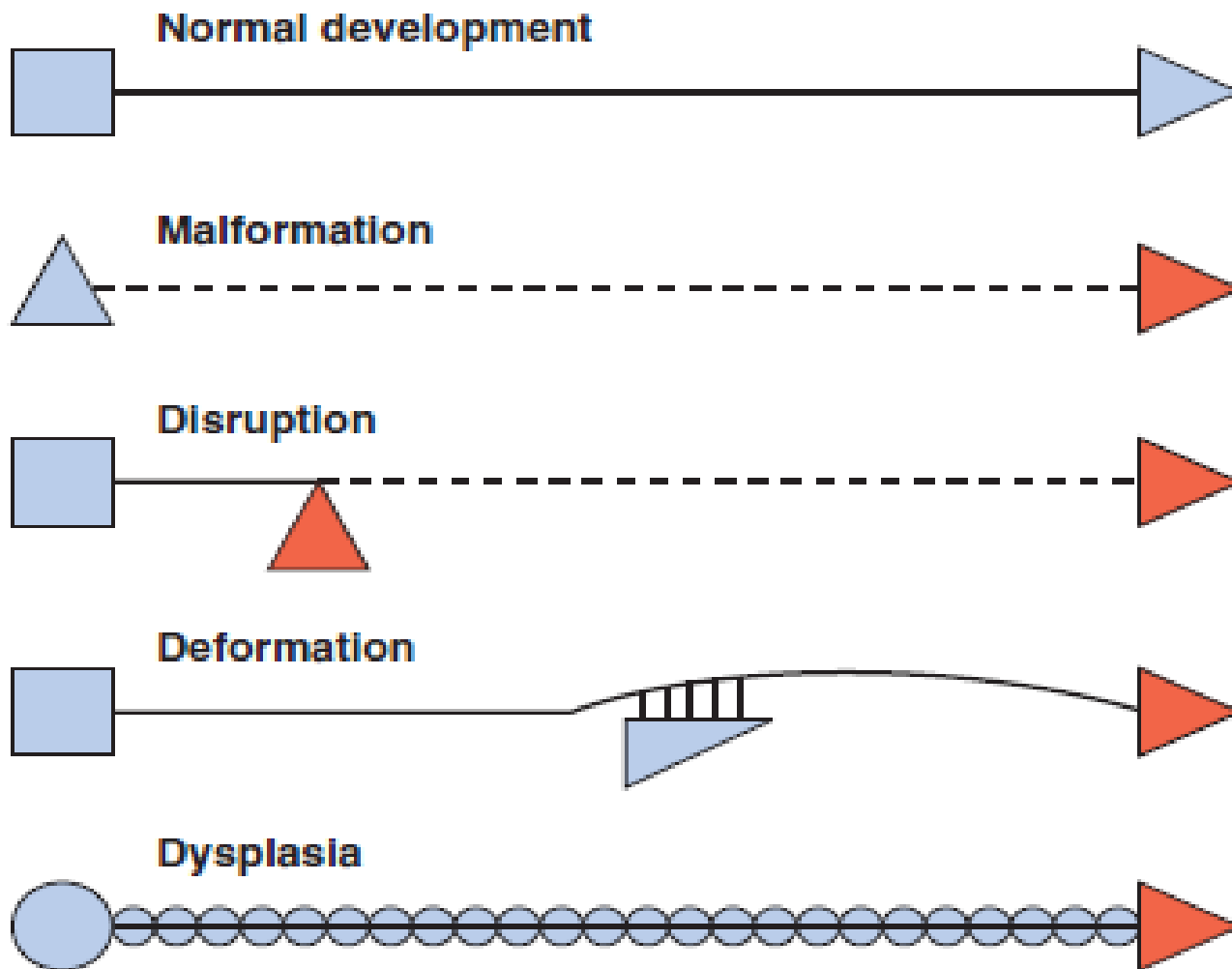
- Abnormal organization of cells into tissue.
- The effects are usually seen wherever that particular tissue is present.
- Most dysplasias are caused by **single-gene defects**



A dysplasia is a disorder at the organ level that is due to problems with tissue development.

- One type of tissue
- **Most dysplasias are due to single gene defect**
- **Examples: Skeletal dysplasia**
Ectodermal dysplasia





Multiple abnormalities

- Sequence
- Syndrome
- Association

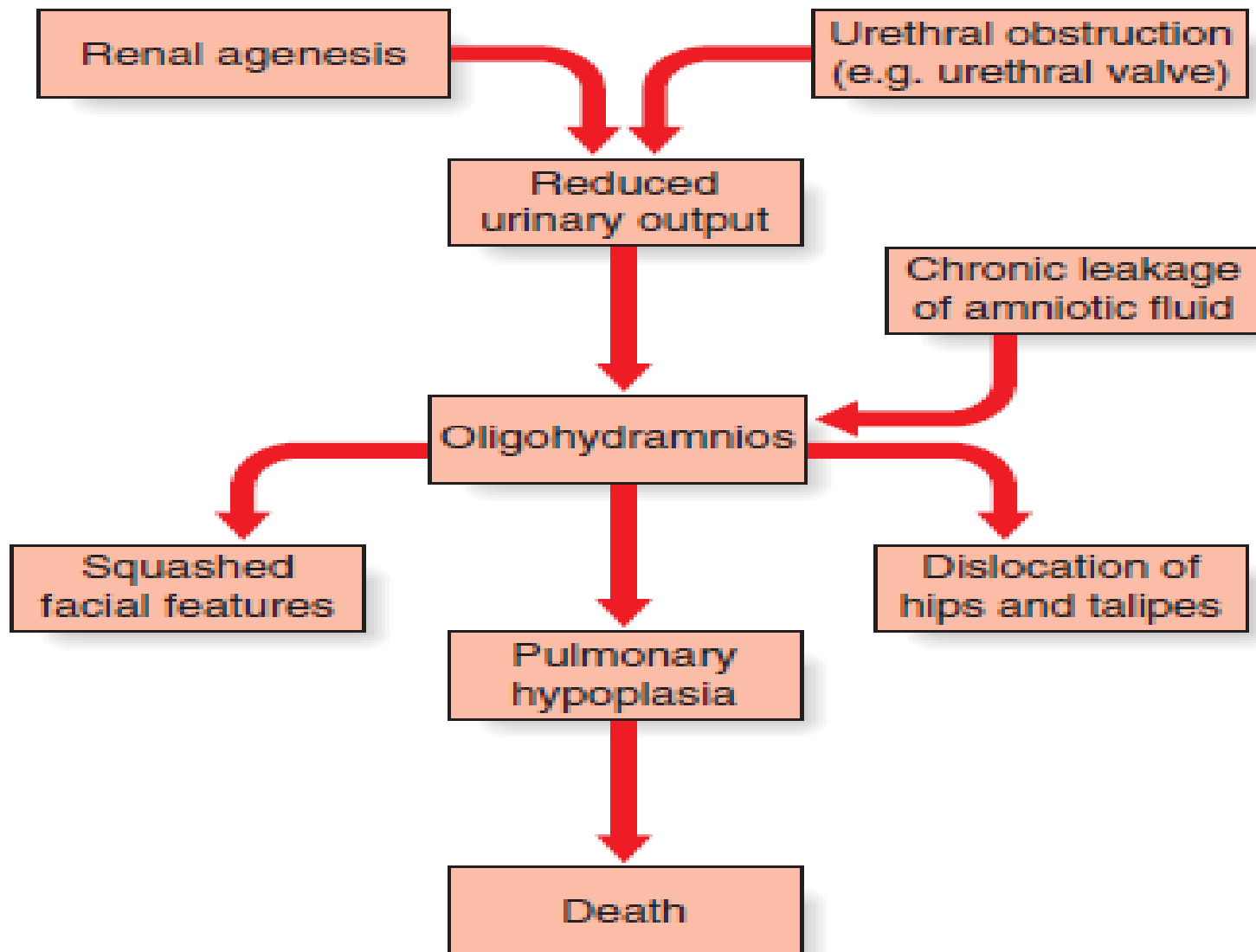
Multiple abnormalities

Sequence

- Findings that occur as a consequence of a **cascade of events** initiated by a single primary factor

- **Example (Potter sequence)**

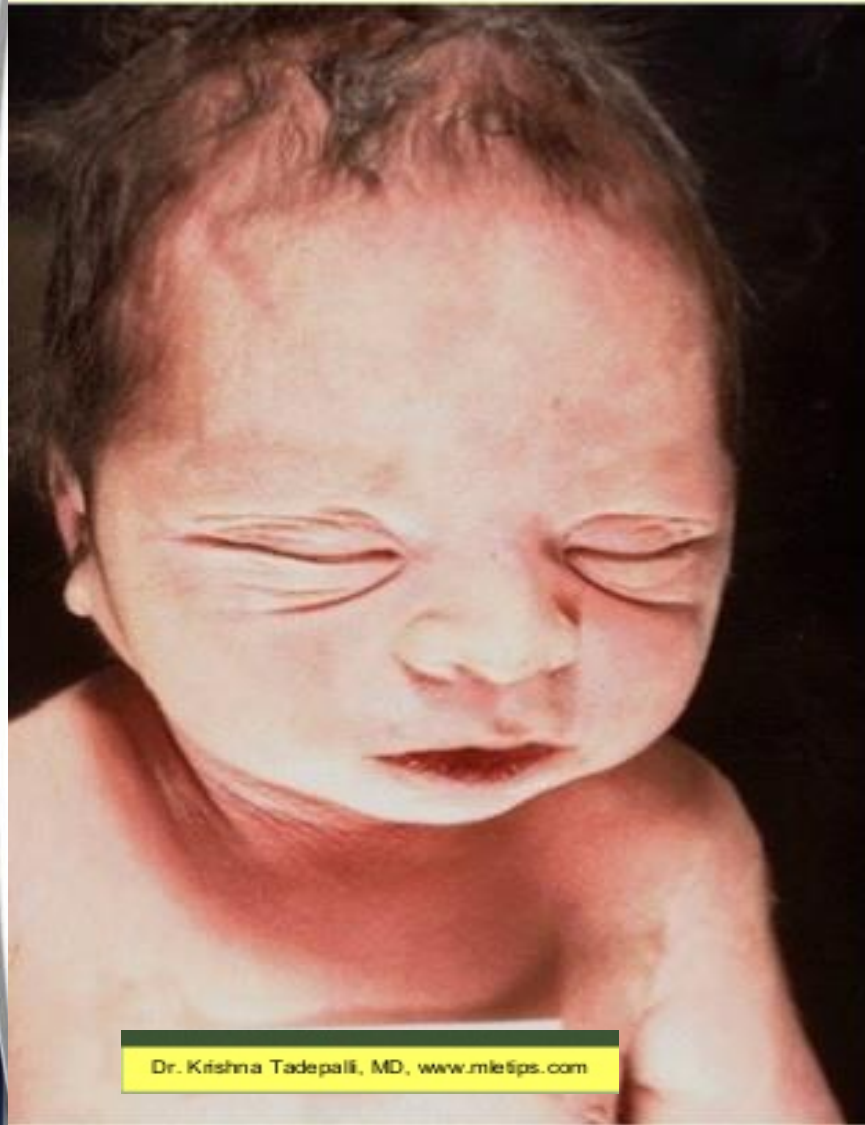
The 'Potter' sequence showing the cascade of events leading to and resulting from oligohydramnios (reduced volume of amniotic fluid).



Potter sequence

- Reduced urinary output
- Oligohydramnios
- Fetal compression
- Squashes facial features, hip dislocation, TEV (club foot), pulmonary hypoplasia
- Death

Potter Sequence



Dr. Krishna Tadepalli, MD, www.mle5ips.com



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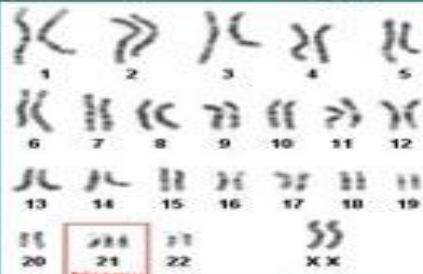
Syndrome

**Recognizable pattern of abnormalities
for which there will be a known
underlying cause**

Example:

- Chromosomal abnormalities e.g. Down syndrome
- Single gene disorder (Van der Woude syndrome)

Down Syndrome Karyotype



flattened nose and face, upward slanting eyes,

single palmer crease, short fifth finger that curves inward

widely separated first and second toes and increased skin creases

Van der Woude syndrome

- ❑ Cleft lip, cleft palate, cleft uvula, lower lip pits and missing teeth(hypodontia)



Association

Certain malformations tend to occur together more often than expected by chance

Can not be explained on the basis of sequence or syndrome

Example

VATER association (vertebral, anal, tracheo-esophageal, renal and /or radial ray abnormalities)

VATER

V: vertebral anomalies

A: Anal atresia

T: Tracheoesophageal
fistula

E: esophageal fistula

R: Renal abnormality/
radial ray defect



VATER (VACTERL) Association

- V: vertebral
- A: anal anomalies
- C: cardiac
- TE: tracheo-esophageal fistula
- R: renal anomalies
- L: limb anomalies

Reference

- ❑ [https:// images.google.com](https://images.google.com)
- ❑ Google Images. The most comprehensive image search on the web.
- ❑ Emery's Elements of Medical Genetics 15th edition 2017 by Turnpenny and Ellard